

WT : EXPERIMENT 7

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Aim: To develop Arduino code that receives messages via Bluetooth and displays the message on the first line of a 16x2 LCD, while simultaneously showing the temperature on the second line.

Theory:

Arduino programming is based on **C/C++** and uses a simplified structure within the Arduino IDE. It's designed for beginners in electronics and embedded systems, making it easy to write code that directly interacts with sensors, actuators, and other components.

Every Arduino sketch (program) has two main functions:

```
void setup() {  
    // Runs once when the Arduino is powered on or reset  
}  
  
void loop() {  
    // Continuously runs after setup, forming the main logic of the program  
}
```

Pin Configuration

Before interacting with components like LEDs or buttons, pins must be configured using `pinMode()`:

```
int ledPin = 13;  
  
void setup() {  
    pinMode(ledPin, OUTPUT); // Set pin 13 as output  
}
```

Controlling Output

You can control components using digitalWrite():

```
digitalWrite(ledPin, HIGH); // Turns the LED ON  
digitalWrite(ledPin, LOW);  // Turns the LED OFF
```

Reading Input

To detect input from sensors or buttons:

```
int buttonState = digitalRead(2); // Reads HIGH or LOW from pin 2
```

Delay Function

To add timing or create effects like blinking:

```
delay(1000); // Pauses code for 1000 milliseconds (1 second)
```

Serial Communication

Serial communication allows the Arduino to send data to your computer via USB. This is useful for debugging or interfacing with Bluetooth modules like HC-05:

```
void setup() {  
  Serial.begin(9600); // Start serial at 9600 baud rate  
}  
  
void loop() {  
  if (Serial.available()) {  
    char data = Serial.read(); // Read data from Bluetooth/Serial  
    Serial.println(data);      // Echo received data  
  }  
}
```

Code:

```
#include <SoftwareSerial.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <DHT.h>

#define DHTPIN 4    // DHT11 module DATA pin connected to D4
#define DHTTYPE DHT11 // DHT11 sensor type

SoftwareSerial BT(2, 3); // RX, TX for Bluetooth
LiquidCrystal_I2C lcd(0x27, 16, 2); // LCD address 0x27 (change to 0x3F if needed)
DHT dht(DHTPIN, DHTTYPE); // Initialize DHT sensor

String lastMessage = "";

void setup() {
  Serial.begin(9600);
  BT.begin(9600);
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("BT Notice Board");
  delay(3000);
  dht.begin(); // Start DHT11 sensor
  delay(2000); // Give sensor time to stabilize
}

void loop() {
  if (BT.available()) {
    lastMessage = ""; // Clear previous message
    while (BT.available()) {
      char c = BT.read();
      lastMessage += c;
      delay(50); // Ensures complete message is received
    }
    Serial.println("Received: " + lastMessage); // Debugging log
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print(lastMessage.substring(0, 16));
  }

  // Read temperature from DHT11 module
  float temp = dht.readTemperature(false) / 2; // Read in Celsius
```

```
// Debugging log
Serial.print("Temperature: ");
Serial.println(temp);

lcd.setCursor(0, 1);
if (isnan(temp)) {
    lcd.print("Temp: Error");
} else {
    lcd.print("Temp: ");
    lcd.print(temp, 2);
    lcd.print(" C ");
}

delay(2500);
}
```